

The big picture

Guiding question(s)

What can be deduced from the temperature change that accompanies chemical or physical change?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

What a great time to be born! What a great time to be alive! Because that generation gets to essentially completely change the world.

Quote from Paul Hawken  (<https://paulhawken.com/>) an environmentalist, entrepreneur, author and activist from the USA

It might not always feel like it, but people in modern societies have longer, healthier lives with more opportunities to work towards happiness than any era ever before. Many developments were essential for us to get here, like the invention of writing, farming and even spoken language, but earlier and perhaps even more important as any of these advancements was the ability to recognise the interconnectedness between changes in temperature and chemical changes of edible and material goods.

Indigenous people have long used energy transfer to prepare foods such as fish through methods of heating directly in a fire, drying in the sun and steaming in a ground oven. Māori *hāngī* refers to the traditional method of cooking food on top of heated rocks buried underground by the Māori people in New Zealand. Long before modern science was developed, it was known, intuitively, that the interactions with heat transformed the fish into something safe to eat for sustenance. This Indigenous knowledge has been essential in the evolution of our species. Some of these methods of fish preparation are shown in **Interactive 1**.

Interactive 1. Indigenous Methods to Prepare Foods.

🕒 More information for interactive 1

The ability of humans to be able to use energy transfers to prepare food, create materials for infrastructure, design diagnostic tests to monitor health and create transportation methods to allow travel all over the world are some examples of reasons today's societies live longer, healthier lives.

Prior learning

Before you study this subtopic, make sure that you understand the following:

- Convert between values in the Celsius and Kelvin scales (see [section S1.1.3a](#)  (/study/app/preview-p/sid-426-cid-753301/book/temperature-and-kinetic-energy-id-43068/)).
- Solve problems involving the relationships between the number of particles, the amount of substance in moles and the mass in grams (see [section S1.4.3](#)  (/study/app/preview-p/sid-426-cid-753301/book/molar-mass-m-id-44262/)).
- Solve problems involving the molar concentration, amount of solute and volume of solution (see [section S1.4.5a](#)  (/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/)).

Practical skills

Once you have completed this subtopic, apply your knowledge of energy transfers by going to [Practical 5a: Determining enthalpy changes in aqueous solutions \(/study/app/preview-p/sid-426-cid-753301/book/determining-enthalpy-changes-in-aqueous-solutions-5a-id-46719/\)](#) and [Practical 5b: Determining enthalpy changes for combustion reactions \(/study/app/preview-p/sid-426-cid-753301/book/determining-enthalpy-changes-for-combustion-reactions-5b-id-46720/\)](#).